

Project 3 Proposal: Flow Around an Actuator Disk Model for UAV Wind Sensor Placement

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I. PROBLEM DESCRIPTION

In multirotor drone design, accurate wind sensing is essential for flight stabilization and environmental monitoring, but propeller-induced flow can corrupt measurements from onboard wind sensors. The present study simulates the two-dimensional incompressible flow generated by a hovering rotor using an actuator-disk model. The propeller is modeled by a thin band of body force that injects momentum into the fluid. The flow is governed by the incompressible Navier–Stokes equations,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

where $\mathbf{f}$ is the actuator-disk body force and $Re = v_h D / \nu$ is the Reynolds number formed from the induced velocity $v_h = \sqrt{T / (2\rho A)}$, the disk diameter D , and the kinematic viscosity ν . Here T is the rotor thrust and $A = \pi D^2 / 4$ is the disk area. Target Reynolds numbers of $Re \in [100, 1000]$ will be explored.

II. NUMERICAL APPROACH

The solver extends the fractional-step (projection) method developed in Project 2 on a staggered grid. The only structural modification is the addition of the actuator-disk body force $\mathbf{f}$ to the momentum predictor step:

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \left[-A(\mathbf{u}^n) + \frac{1}{Re} L \mathbf{u}^n + \mathbf{f} \right], \quad (2)$$

where A is the discrete advection operator and L is the vector Laplacian. The disk force is applied uniformly over a single column of u -face cells spanning the rotor diameter, yielding a spatially compact forcing that is straightforward to implement within the existing sparse-matrix framework. The pressure correction step and divergence-free projection remain identical to Project 2. Second-order central differences are used on a uniform Cartesian grid with explicit Euler time integration ($CFL \leq 0.5$), and a scipy CG solver handles the pressure Poisson system at each step.

III. PLANNED FEATURES AND AI INTEGRATION

Actuator-disk body force module. Implement a sparse forcing vector that injects a prescribed thrust coefficient $C_T = T / (\frac{1}{2} \rho v_h^2 A)$ into the u -momentum equation across the disk region, with AI assistance to verify the index mapping from the 2D staggered grid to the flattened 1D system.

Wake velocity post-processing pipeline. Build routines to extract centerline velocity profiles, compute thrust from integrated momentum flux, and compare against theory, with AI assistance for the quadrature and plotting.

Grid-convergence and time-step study. Automate a sweep over grid resolutions (32^2 to 256^2), log the L^2 velocity error relative to the analytical hover solution, and produce convergence-rate plots, using AI to scaffold the sweep and curve-fitting.

Sensor-placement analysis. Map $|\mathbf{u} - \mathbf{u}_\infty|$ across the domain to identify regions of low flow disturbance suitable for anemometer placement, with AI assistance for contouring and thresholding.

IV. VALIDATION PLAN

The primary benchmark is Froude–Rankine actuator-disk momentum theory, which predicts that the stream-wise velocity at the disk plane equals the hover-induced velocity v_h , and that the far-wake velocity approaches $2v_h$. The simulation will be validated by comparing (i) the centerline velocity one diameter downstream against the theoretical value $2v_h$, (ii) the integrated thrust extracted from the momentum flux against the prescribed forcing, and (iii) the radial velocity profile in the far wake against the top-hat actuator-disk prediction.

V. EXPECTED OUTCOMES AND TIMELINE

- **Week 1 (Apr. 24 – May 1):** Integrate body force term into Project 2 solver. verify steady hover solution for a single test case at $Re = 200$. Quantitative validation against momentum theory. grid- and time-step convergence study. confirm second-order spatial accuracy.
- **Week 2 (May 1 – May 6):** Parameter study over $Re \in [100, 1000]$; sensor-placement post-processing. generate all figures. Draft final report. compile AI Usage Log. Submit.